

CHAIYAPHUM, Thailand

WMO: 484030

Lat: 15.812N

Lon: 102.029E

Elev: 184

StdP: 99.13

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.8	17.3	8.2	6.9	20.9	10.1	7.9	21.3	5.3	25.3	4.9	25.5	0.6	50	0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.4	38.3	25.3	37.2	25.0	36.1	24.8	27.5	34.2	26.8	32.8	26.5	32.2	1.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.7	29.9	25.5	21.2	29.2	25.2	20.7	28.6	88.4	34.3	85.6	33.0	84.0	32.3	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.6	3.9	3.3	13.0	39.7	2.0	1.1	11.6	40.5	10.4	41.2	9.3	41.8	7.9	42.6		
(5)			10.6	28.9	1.8	1.0	9.3	29.6	8.3	30.1	7.3	30.7	6.0	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	25.0	27.2	29.4	30.7	29.9	29.4	28.6	28.2	28.0	27.8	26.6	24.8		
	DBStd	2.61	2.38	2.66	2.47	2.05	1.80	1.34	1.38	1.17	1.14	1.30	1.96	2.59		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1		
	CDD10.0	6556	466	481	602	621	616	581	577	564	541	551	497	460		
	CDD18.3	3515	208	248	343	371	357	331	319	306	291	292	247	202		
	CDH23.3	39104	1908	2737	4318	4846	4352	3973	3549	3232	2985	3027	2383	1795		
	CDH26.7	17556	772	1361	2355	2699	2124	1798	1443	1199	1057	1141	940	666		
(14) Wind	WSAvg	0.9	0.9	0.9	0.9	0.8	0.8	1.0	1.1	1.0	0.7	0.9	1.0	1.1		
(15) Precipitation	PrecAvg	1207	4	18	50	89	166	141	151	140	285	139	20	5		
	PrecMax	1670	22	96	105	175	257	238	314	276	620	366	109	36		
	PrecMin	811	0	0	0	40	7	30	62	47	104	16	0	0		
	PrecStd	256	8	25	30	41	75	61	78	58	148	87	29	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	37.4	39.2	40.2	38.7	36.7	36.3	34.6	34.1	34.0	35.0	34.2		
		MCWB	22.4	23.1	24.3	26.3	27.5	25.9	25.7	25.7	26.0	25.5	24.2	23.4		
	2%	DB	33.5	36.2	37.9	38.8	37.0	35.2	34.4	33.4	33.0	33.0	33.6	32.8		
		MCWB	22.5	23.0	24.1	25.7	26.3	25.8	25.5	25.6	25.9	25.0	24.2	22.9		
	5%	DB	32.2	35.0	36.8	37.4	35.5	34.2	33.2	32.5	32.2	32.2	32.3	31.6		
		MCWB	22.3	23.1	24.1	25.4	26.2	25.7	25.4	25.6	25.7	24.7	23.8	22.5		
	10%	DB	30.9	33.5	35.5	36.1	34.2	33.1	32.1	31.5	31.3	31.5	31.2	30.3		
		MCWB	21.8	22.9	24.1	25.3	26.0	25.7	25.3	25.4	25.5	24.6	23.3	22.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.5	27.3	28.2	28.9	27.3	27.0	27.1	27.1	26.8	26.0	24.9		
		MCDB	31.6	33.9	35.4	38.3	36.2	33.4	32.7	31.5	31.7	31.6	31.3	31.0		
	2%	WB	24.1	24.7	26.0	26.9	27.7	26.7	26.4	26.5	26.6	26.2	25.4	24.1		
		MCDB	30.6	32.8	34.3	35.5	34.4	32.4	31.6	30.9	30.9	30.6	30.6	30.3		
	5%	WB	23.3	24.2	25.3	26.2	26.9	26.3	26.0	26.1	26.2	25.8	24.9	23.3		
		MCDB	29.7	32.0	33.5	34.4	33.0	31.8	31.0	30.6	30.3	30.0	30.0	29.7		
	10%	WB	22.6	23.6	24.8	25.8	26.3	26.0	25.7	25.7	25.9	25.4	24.3	22.6		
		MCDB	28.9	31.1	32.6	33.5	32.0	31.2	30.3	30.0	29.8	29.4	29.0	29.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.9	11.3	10.4	8.4	7.7	7.2	6.9	6.7	7.3	8.8	10.2		
		MCDBR	12.9	13.4	13.1	12.0	10.2	9.0	8.6	8.2	7.9	8.2	9.9	11.2		
		MCWBR	4.3	3.8	3.6	3.1	2.7	2.1	2.2	2.3	2.3	2.4	3.0	3.5		
	5% WB	MCDBR	10.9	11.7	11.7	10.8	9.2	8.1	7.5	7.4	7.2	7.3	8.6	10.0		
		MCWBR	4.1	3.6	3.8	3.3	3.1	2.4	2.5	2.5	2.4	2.4	2.9	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.488	0.593	0.654	0.606	0.507	0.458	0.446	0.454	0.497	0.540	0.479	0.472		
	taud		2.052	1.804	1.704	1.856	2.170	2.322	2.355	2.319	2.184	2.026	2.169	2.127		
	Ebn at Noon		798	731	702	738	803	835	846	848	814	767	802	801		
	Edh at Noon		162	217	246	211	152	129	125	131	150	171	143	147		
(44) All-Sky Solar Radiation	RadAvg		4.87	5.23	5.36	5.77	5.55	5.24	4.93	4.78	4.67	4.86	4.91	4.77		
	RadStd		0.31	0.23	0.27	0.24	0.31	0.28	0.31	0.29	0.27	0.34	0.21	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+158
(47) Regional (26 neighbors)		+0.29	N/A	N/A	N/A	+0.21	+0.20	N/A	N/A	+129	+129

Nomenclature: See separate page